2nd Apr 2026 Shift 2 (Actual Paper)

Mathematics Section A

1. Let α, β be the roots of the equation $x^2 - 3x + \mathrm{r} = 0$, and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 + 3x + \mathrm{r} = 0$. If the roots of the equation $x^2 + 6x = \mathrm{m}$ are $2\alpha + \beta + 2\mathrm{r}$ and $\alpha - 2\beta - \frac{\mathrm{r}}{2}$, then m is equal to :

(A) –135 (B) –567

(C) 135 (D) 567

2. Let the circles $C_1 : |z| = r$ and $C_2 : |z - 3 - 4i| = 5, z \in \mathrm{C}$, be such that C_2 lies within C_1. If z_1 moves on C_1, z_2 moves on C_2 and $\min|z_1 - z_2| = 2$, then $\max|z_1 - z_2|$ is equal to:

(A) 12 (B) 17 (C) 22 (D) 24

3. If the system of equations

$x + 5y + 6z = 4$

$2x + 3y + 4z = 7$

$x + 6y + az = \mathrm{b}$

has infinitely many solutions, then the point (a, b) lies on the line

(A) y-x=3 (B) x-y=3 (C) x+y=11 (D) x+y=12

4. Let $a_1, a_2, a_3, \ldots$ be an A.P. and $g_1 = a_1, g_2, g_3, \ldots$ be an increasing G.P. If $a_1 = a_2 + g_2 = 1$ and $a_3 + g_3 = 4$, then $\mathrm{a}_{10} + \mathrm{g}_5$ is equal to :

(A) 81 (B) 76 (C) 62 (D) 55

5. The sum $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \cdots$ up to 8 terms, is :

(A) 70 (B) 71 (C) 72 (D) 73

6. If for $3 \leqslant \mathrm{r} \leqslant 30$, $({}^{30}\mathrm{C}_{30-\mathrm{r}}) + 3\,({}^{30}\mathrm{C}_{31-\mathrm{r}}) + 3\,({}^{30}\mathrm{C}_{32-\mathrm{r}}) + ({}^{30}\mathrm{C}_{33-\mathrm{r}}) = {}^{\mathrm{m}}\mathrm{C}_{\mathrm{r}}$, then m equals:

(A) 31 (B) 32 (C) 33 (D) 34

7. Let $\mathrm{p_n}$ denote the total number of triangles formed by joining the vertices of an n -side regular polygon. If $\mathrm{p_{n+1}} - \mathrm{p_n} = 66$, then the sum of all distinct prime divisors of n is :

(A) 7 (B) 8 (C) 5 (D) 6

8. A man throws a fair coin repeatedly. He gets 10 points for each head he throws and 5 points for each tail he throws. If the probability that he gets exactly 30 points is $\frac{\mathrm{m}}{\mathrm{n}}, \gcd(\mathrm{m}, \mathrm{n}) = 1$, then $\mathrm{m} + \mathrm{n}$ is equal to:

(A) 53 (B) 55 (C) 107 (D) 105

9. The mean and variance of n observations are 8 and 16 , respectively. If the sum of the first ($n - 1$) observations is 48 and the sum of squares of the first ($n - 1$) observations is 496 , then the value of n is :

(A) 21 (B) 16 (C) 13 (D) 7

10. Let a circle pass through the origin and its centre be th point of intersection of two mutually perpendicular line $x + (\mathrm{k} - 1)y + 3 = 0$ and $2x + \mathrm{k}^2 y - 4 = 0$. If the lin $x - y + 2 = 0$ intersects the circle at the points A and E then $(AB)^2$ is equal to :

(A) 10 (B) 27 (C) 18 (D) 34

11. Let O be the origin, and P and Q be two points on th rectangular hyperbola $xy = 12$ such that the mid point o the line segment PQ is $\left(\frac{1}{2}, -\frac{1}{2}\right)$. Then the area of the tri angle OPQ equals :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$

(C) $\frac{7}{2}$ (D) $\frac{9}{2}$

12. Let the parabola $y = x^2 + px + q$ passing through th point $(1, -1)$ be such that the distance between its verte and the x-axis is minimum. Then the value of $\mathrm{p}^2 + \mathrm{q}^2$ is :

(A) 2 (B) 4 (C) 5 (D) 8

13. Let $P = \{\theta \in [0, 4\pi] : \tan^2\theta \neq 1\}$ and $S = \{a \in \mathbf{Z} : 2\left(\cos^8\theta - \sin^8\theta\right)\sec 2\theta = a^2, \theta \in P\}$. The $n(S)$ is :

(A) 0 (B) 1 (C) 2 (D) 3

14. Let the vectors $\overrightarrow{\mathrm{a}} = -\hat{\imath} + \hat{\jmath} + 3\hat{k}$ and $\overrightarrow{\mathrm{b}} = \hat{\imath} + 3\hat{\jmath} + \hat{k}$ For some $\lambda, \mu \in \mathbf{R}$, let $\overrightarrow{\mathrm{c}} = \lambda\overrightarrow{\mathrm{a}} + \mu\overrightarrow{\mathrm{b}}$. If $\overrightarrow{\mathrm{c}} \cdot (3\hat{\imath} - 6\hat{\jmath} + 2\hat{k}) = 10$ and $\overrightarrow{\mathrm{c}} \cdot (\hat{\imath} + \hat{\jmath} + \hat{k}) = -2$, then $|\overrightarrow{\mathrm{c}}|^2$ is equal tc

(A) 8 (B) 12 (C) 14 (D) 15

15. Let the point A be the foot of perpendicular drawn fror the point $\mathrm{P}(\mathrm{a}, \mathrm{b}, 0)$ on the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-\alpha}{3}$. If th midpoint of the line segment PA is $\left(0, \frac{3}{4}, \frac{-1}{4}\right)$, then th value of $a^2 + b^2 + \alpha^2$ is equal to :

(A) 1 (B) 2 (C) 6 (D) 9

16. Two adjacent sides of a parallelogram PQRS are given b $\overrightarrow{\mathrm{PQ}} = \hat{\jmath} + \hat{k}$ and $\overrightarrow{\mathrm{PS}} = \hat{\imath} - \hat{\jmath}$. If the side PS is rotate about the point P by an acute angle α in the plane of th parallelogram so that it becomes perpendicular to the sid PQ , then $\sin^2\left(\frac{5\alpha}{2}\right) - \sin^2\left(\frac{\alpha}{2}\right)$ is equal to:

(A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{3}}{4}$ (D) $\frac{2\sqrt{3}}{5}$

17. The value of $\int_0^{20\pi}\left(\sin^4 x + \cos^4 x\right) dx$ is equal to :

(A) $\frac{15\pi}{2}$ (B) 25π

(C) 15π (D) $\frac{25\pi}{2}$

18. Let $f(x)$ be a polynomial of degree 5 , and have extrema at $x=1$ and $x=-1$. If $\lim_{x\to0}\left(\frac{f(x)}{x^3}\right)=-5$, then $f(2)-f(-2)$ is equal to :

(A) 0 (B) 50 (C) 92 (D) 112

19. Let $f(x)=\int\left(\frac{16x+24}{x^2+2x-15}\right)\mathrm{d}x$. If $f(4)=14\log_e(3)$ and $f(7)=\log_e\left(2^\alpha\cdot3^\beta\right),\alpha,\beta\in\mathbf{N}$, then $\alpha+\beta$ is equal to:

(A) 31 (B) 37 (C) 39 (D) 41

20. Let $x=x(y)$ be the solution of the differential equatio $2y^2\frac{\mathrm{d}x}{\mathrm{d}y}-2xy+x^2=0, y>1, x(\mathrm{e})=\mathrm{e}$. Then $x(\mathrm{e}^2)$ i equal to :

(A) $\frac{3}{2}e^2$ (B) $\frac{2}{3}e^2$

(C) e^2 (D) $2e^2$

Mathematics Section B

21. Let $\mathrm{A}=\{2,3,4,5,6\}$. Let R be a relation on the set $\mathrm{A}\times\mathrm{A}$ given by $(x,y)\mathrm{R}(z,w)$ if and only if x divides z and $y\leqslant w$. Then the number of elements in R is ____ .

22. Consider the matrices $\mathrm{A}=\left[\begin{array}{cc}2&-2\\4&-2\end{array}\right]$ and $\mathrm{B}=\left[\begin{array}{cc}3&9\\1&3\end{array}\right]$. If matrices P and Q are such that $\mathrm{PA}=\mathrm{B}$ and $AQ=B$, then the absolute value of the sum of the diagonal elements of $2(P+Q)$ is ____ .

23. Let A be the point $(3,0)$ and circles with variable diameter AB touch the circle $x^2+y^2=36$ internally. Let th curve C be the locus of the point B. If the eccentricity o C is e, then $72e^2$ is equal to ____ .

24. If the area of the region bounded by $16x^2-9y^2=14$ and $8x-3y=24$ is A , then $3(\mathrm{A}+6\log_e(3))$ is equa to ____ .

25. The number of points in the interval $[2,4]$, at which th function $f(x)=\left[x^2-x-\frac{1}{2}\right]$, where $[\cdot]$ denotes the greatest integer function, is discontinuous, is ____。

Physics Section A

26. Dimensions of universal gravitational constant (G) in terms of Planck's constant (h), distance (L), mass (M) and time (T) are ____ .

(A) $[\mathrm{hTLM}^{-2}]$

(B) $[\mathrm{hT}^{-1}\mathrm{LM}^{-2}]$

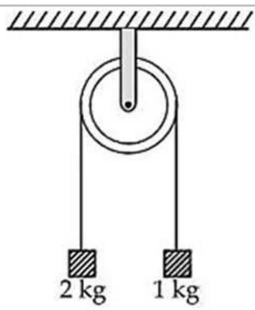
(C) $[\mathrm{hTL}^2\mathrm{M}^{-2}]$

(D) $[\mathrm{h}^{-1}\,\mathrm{T}^{-1}\mathrm{LM}^{-2}]$

27. A 0.5 kg mass is in contact against the inner wall of a cylindrical drum of radius 4 m rotating about its vertical axis. The minimum rotational speed of the drum to enable the mass to remain stuck to the wall (without falling) is $5\mathrm{rad/s}$. The coefficient of friction between the drum's inner wall surface and mass is ____ . (Take $g=10\ \mathrm{m/s^2}$)

(A) 0.1 (B) 0.5 (C) 0.7 (D) 0.3

28. Two blocks of masses 2 kg and 1 kg respectively, are tied to the ends of a string which passes over a light frictionless pulley as shown in the figure below. The masses are held at rest at the same horizontal level and then released. The distance traversed by the centre of mass in 2 s is ____ m. (Take $g=10\ \mathrm{m/s^2}$)



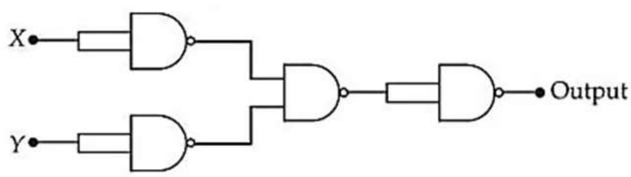
(A) 3.33 (B) 3.12 (C) 2.22 (D) 1.42

29. A particle having charge $10^{-9}\mathrm{C}$ moving in $x-y$ plane i fields of $0.4\hat{j}\ \mathrm{N/C}$ and $4\times10^{-3}\hat{k}\ \mathrm{T}$ experiences a forc of $(4\hat{i}+2\hat{j})\times10^{-10}\ \mathrm{N}$. The velocity of the particle a that instant is ____ $\mathrm{m/s}$.

(A) $50\hat{i}+100\hat{j}$ (B) $100\hat{i}+50\hat{j}$

(C) $-50\hat{i}+100\hat{j}$ (D) $50\hat{i}-100\hat{j}$

30. If X and Y are the inputs, the given circuit works a ____。



(A) OR gate (B) AND gate (C) NAND gate (D) NOR gat